

Spectral Relativity Theory — Version 3

Appendix: Status of the Central Object — Internal Coherence and External Validation

P. Guiffra

August 2026 (revision of the July 2026 appendix)

Revision note

This appendix supersedes the July 2026 version (*“Rigidity Observation of the Central Object”*). An internal audit, conducted under the project’s anti-circularity protocol, showed that the central claim of the earlier version — *“14 frameworks, 14 independent reconstructions, no failures”* — does not survive strict scrutiny. Most of the mathematical frameworks cited already presuppose, at some stage of their construction, the kink shape $\Phi = \tanh$ or the quartic potential V_2 . They are therefore not independent reconstructions of the quartic itself.

Two clearly different levels must nonetheless be distinguished among these frameworks, rather than treated as a single block:

- frameworks that presuppose **only** V_2 and **genuinely derive** $\Phi = \tanh$ from it through a non-trivial calculation (Section 2.1);
- frameworks that presuppose $\Phi = \tanh$ **already built** and merely compute properties of it (Section 2.2) — these add essentially nothing new once the first group is established.

Two frameworks only constitute genuine, **independent external** validation, against measured, unadjusted data: comparison with lattice QCD data and comparison with direct turbulence measurements (Section 3). It is this latter category that provides the theory’s most significant reinforcement, precisely because it exposes a quantitative prediction to genuine risk of failure — unlike internal-coherence checks, which cannot structurally fail.

1 The Central Object

The central object admits three equivalent representations:

$$\Phi(x) = \tanh(\alpha x) \quad [\text{kink profile}] \quad (1)$$

$$\rho(x) = \alpha \operatorname{sech}^2(\alpha x) \quad [\text{spectral density}] \quad (2)$$

$$\psi_0(x) = \operatorname{sech}^2(\alpha x) \quad [\text{Goldstone mode}] \quad (3)$$

related by $\rho = \Phi'$ and $\psi_0 = \rho/\alpha$.

2 Part A — Internal Coherence (Eleven Frameworks)

2.1 Genuine conditional derivations (at fixed V_2)

These two frameworks presuppose only the choice of the quartic potential V_2 — not the final shape of Φ — and establish $\Phi = \tanh$ through a non-trivial mathematical result. They remain **conditional** (they do not select V_2 itself among admissible potentials — see Section 4), but they are genuine derivations, not tautologies.

A1. Calculus of variations. Theorem 1.1: at fixed potential V_2 , $\Phi = \tanh$ is the unique monotone, finite-action minimizer of the symmetric double-well system.

A2. Integrable ODE / Bogomolny. The Bogomolny equation of the quartic potential, $\Phi' = \alpha(1 - \Phi^2)$, integrates by elementary quadrature and yields $\Phi = \tanh$ through explicit resolution — a genuine calculation, not a substitution.

2.2 Immediate consequences (with $\Phi = \tanh$ already built)

These eight frameworks start from the $\tanh$ kink *already obtained* in A1–A2 and compute properties of it. They confirm the object’s internal coherence across eight different mathematical languages — a non-trivial fact in itself — but add no new derivation: once A1 or A2 is established, each of them follows directly.

A3. Spectral theory / Pöschl–Teller. $L = -d^2/dx^2 + 4\alpha^2 - 6\alpha^2 \operatorname{sech}^2(\alpha x)$: the Hessian of $\mathcal{A}[\Phi]$ evaluated on the already-constructed kink.

A4. Sturm–Liouville. Inherits from A3: the operator L is already the Pöschl–Teller operator.

A5. Morse theory. Morse index of the operator L already constructed in A3–A4.

- A6. Differential geometry (phase plane).** The unique heteroclinic orbit from $(-1, 0)$ to $(+1, 0)$ is that of the Bogomolny equation already solved in A2.
- A7. Pohozaev identities.** Integrals computed *on* the already-known tanh trajectory — $J_2 = 6/5$ is a property of the kink once built.
- A8. Dynamical systems.** Uniqueness via Cauchy–Lipschitz applies to the Bogomolny ODE already solved in A2.
- A9. Spectral Riemannian geometry.** The flatness of $ds^2 = d\omega^2/\rho(\omega)$ is a solid *local* result, independently established (vanishing order $\theta = 2$, see main text, Theorem 4.1). Extending this local flatness to the *global* sech^2 shape is not established, however — this point remains open.
- A10. Pure invariants.** $J_1 = 6$, $J_2 = 6/5$, $J_3 = 1/\sqrt{3}$ are *computed from the already-known kink shape*. Universal once tanh is posited.

2.3 Status pending verification

- A11. KPP / Reaction-diffusion.** The fixed point $\rho^* = (3\lambda/2\gamma) \text{sech}^2(\alpha x)$ requires further verification to establish whether it is a genuine consequence of the coupled flow without a prior shape assumption (in which case it would belong with Section 2.1), or an ansatz verified a posteriori (in which case it would belong with Section 2.2). *Status undecided at this stage.*

Withdrawn framework

The “*Darboux factorization / SUSY-QM*” framework of the previous version claimed that SUSY reconstruction from a zero mode necessarily yields $\psi_0 = \text{sech}^2$, $\Phi = \tanh$, quartic V . A literature check (inverse scattering theory, reflectionless potentials with N bound states) shows this claim is **incorrect**: the potential $-n(n+1) \text{sech}^2(\kappa x)$ is a particular case — the most symmetric one — of a broader family of reflectionless potentials with n bound states, parametrized by independent scale and translation parameters. The generic SUSY reconstruction does not select the quartic. This framework is withdrawn.

3 Part B — Independent External Validations

Only these two frameworks confront an RST prediction with independently measured data, without adjustment to produce agreement. This category constitutes the theory’s most significant reinforcement.

B1. Yang–Mills / QCD

String tension $\sigma = 0.194 \text{ GeV}^2$ (deviation $< 0.5\%$ vs. lattice), critical temperature $T_c = 156.7 \text{ MeV}$ (deviation 1.1% vs. lattice).

B2. Navier–Stokes / Turbulence — updated August 2026

Spectral flux $\Pi(k)$, sigmoid coefficient α_Π , compared against the JHTDB database.

Dataset	Re_λ	Deviation vs. reference
isotropic1024coarse	≈ 433	— (reference, $\alpha_\Pi = 0.9605$)
isotropic4096	≈ 610	1.39%
isotropic8192	≈ 1257	0.22%

Adopted formulation: *the predicted value of α_Π remains consistent with the reference to better than 5% over the tested range ($Re_\lambda \approx 433$ to 1257, a factor of ~ 3). No systematic drift with Reynolds number is observed.* Numerical consistency check: Parseval ratio = 1.000 on the isotropic8192 extraction.

The significance of this result lies in its nature: RST provides a precise quantitative prediction, exposed to genuine risk of falsification against independent data, at a turbulent Reynolds number nearly three times the reference point — and the prediction holds, without retuning the model.

4 On the Status of the Quarticity Axiom $\mathcal{A}_Q$

A separate audit, conducted the same month, tested five mechanisms internal to RST that could potentially select degree 4 without presupposing the sought result: elementary Bogomolny quadrature, exactly two bound states, BPS/SUSY factorization, extremal kink tension, and the presence of an integrable (Lax) structure in the KPP system. **None of these five mechanisms selects the quartic.** A sixth criterion — strict reflectionlessness $R(k) \equiv 0$ — appears numerically to select $A = 0$ within the tested family $V_{A,B} = (1 - \Phi^2)^2(A\Phi^2 + B)$, but this result remains numerical, not analytically proven, and its own internal justification within RST remains itself an open question.

The formulation adopted for the theory is therefore:

The KPP construction fixes the universal spectral attractor ($\Theta \rightarrow -1/2$). The variational reconstruction of the Pöschl–Teller $n = 2$ sector additionally requires the independent quarticity axiom $\mathcal{A}_Q$, whose derivation from the KPP/BPS sector is not established to date.

5 Conclusion

Of the fourteen frameworks in the previous version: two establish a genuine conditional derivation of $\tanh$ at fixed V_2 (Section 2.1), eight are immediate consequences thereof (Section 2.2), one has a status still undecided, one was withdrawn for factual error, and two constitute genuinely independent external validation (Part B), now reinforced by an additional data point at $Re_\lambda \approx 1257$.

This reformulation does not diminish the value of the work accomplished: the conditional derivation in A1–A2, the coherence across eight mathematical languages in A3–A10, and the two external validations holding without retuning over a very wide physical range, are three real results of different natures. Distinguishing them explicitly — rather than presenting them as a single, uniform “universal mathematical rigidity” — is what the project’s anti-circularity protocol requires, and it is what makes each of these results more credible taken on its own.